# COMP 462 / 561 - Homework #3
# Due on November 14 2024, 23:59 on MyCourses

**Question 1 (80 points).**
In this question, you will implement the Viterbi algorithm and apply it to the gene finding HMM seen in class to predict genes in a bacterial genome.

*Vibrio cholerae* is the bacteria that causes cholera. It was first sequenced in 2000 (see https://www.nature.com/articles/35020000 ). You can download the genome sequence on MyCourses (Vibrio_cholerae.GFC_11.dna.toplevel.fa.gz). Note: although the genome of *Vibrio cholera* consists of two chromosomes, the sequence file contains more than two sequences, because it is not fully assembled.

The gene annotation is available in the file Vibrio_cholerae.GFC_11.37.gff3.gz . The format of both files should be fairly self-explanatory, but you can find more information about the gff3 file format here: http://gmod.org/wiki/GFF3. The file contains annotations of different types of functional elements. We will only consider those called "CDS" (for CoDing Sequences); ignore the other lines (i.e. those labeled "gene" or "exon" or "mRNA").

Important notes:

- One important detail we've omitted in our in-class discussion is that genes can actually be located either on the forward or the reverse DNA strand. In class, we have only considered the genes located on the forward strand. In this assignment, we will also only focus on the genes on the forward strand. It is not much more complicated to handle genes on the negative strand, but it's just annoying, so we will pretend that genes on the negative strand do not exist, i.e. that portions of the genome occupied by genes on the reverse strand are actually intergenic regions.
- Another level of complexity that we will ignore is the (fairly rare) possibility that two genes can overlap each other. This obviously breaks our HMM gene finder. Because such events are very rare, we will assume that this never occurs. The gff3 file given to you actually contains a few of those pairs of overlapping genes. You can deal with them in whichever way you find simplest; this is not going to affect your results significantly.
- In bacteria, there are actually multiple possible start codons (not only ATG).

a) (10 points) Using the genome sequence and gene annotation for *Vibrio cholerae*, infer the (i) average length of intergenic regions, (ii) average length of CDS, (iii) the nucleotide frequency table for intergenic regions; (iv) the codon frequency table for genic regions. Report those four tables.

b) (40 points) Using the programming language of your choice, implement the Viterbi algorithm. Your program does not need to work with an arbitrary HMM. It only needs

to work for the specific bacterial gene-finding HMM seen in class (4 states: Intergenic, Start, Middle, Stop). The program's argument should be:

I. A Fasta file containing one or more bacterial genomic sequences.
II. A configuration file (formatted as you want) containing the emission and transition probabilities for your HMM, which should be based on the information found in (a). Assume that the initial state probability is 1 for the intergenic state, and zero for all other states.

The output should be a GFF3 file with one line per predicted gene (only lines corresponding to portions annotated as genes need to be included), e.g. something like:

```
DN38.contig00011  ena   CDS   85269 86705 .     +     0     .
DN38.contig00011  ena   CDS   87006 87230 .     +     0     .
DN38.contig00011  ena   CDS   88079 89323 .     +     0     .
```

Submit your code, along with a simple explanation about how to run it.

**Hints!** One key challenge in this implementation is dealing with the fact that some HMM states emit a single nucleotide, while others emit triplets of nucleotides. How should the Viterbi algorithm be adjusted? First make sure that you are absolutely clear and precise about what is the meaning of the probability stored at entry V(s,i) of your dynamic programming table. If this isn't 100% clear in your head, your implementation will likely not work. Second, keep in mind that although the original Viterbi algorithm computes V(s,i) as a function of V(•,i-1), nothing prevents you from changing that, e.g. compute V(s,i) based on V(•,i-1) or V(•,i-3), depending on whether state s emits one or three nucleotides.

c) (10 points) *Vibrio vulnificus* is a species of bacteria closely related to *Vibrio cholerae*. In 2005, it caused an outbreak following the hurricane Katrina (https://www.ncbi.nlm.nih.gov/pubmed/16195696 ). Its genome was sequenced; the sequence is in the file Vibrio_vulnificus.ASM74310v1.dna.toplevel.fa.gz

Run your program on this genome, using the parameters obtained for *Vibrio cholerae*. Submit the GFF3 file with your gene predictions (forward strand only).

d) (10 points) The genes of *Vibrio vulnificus* were annotated based on a variety of types of evidence. The gene annotation is in the file Vibrio_vulnificus.ASM74310v1.37.gff3.gz

Evaluate the accuracy of your predictions against this gene annotation (always limiting our attention to genes on the positive strand). Specifically, report:

- The fraction of annotated genes on the positive strand that:

    - Perfectly match both ends of one of your predicted genes
    - Match the start but not the end of a predicted gene
    - Match the end but not the start of a predicted gene
    - Do not match neither the start not the end of a predicted gene
- The fraction of your predicted genes that:
    - Perfectly match both ends of one of an annotated genes
    - Match the start but not the end of an annotated gene
    - Match the end but not the start of an annotated gene
    - Do not match neither the start not the end of an annotated gene

Note: Gene prediction is not an easy task! You should expect that only approximately half of the annotated genes are perfectly predicted.

e) (10 points) What properties of annotated genes are associated to an elevated risk of being partially or completely missed by your predictor? What are the properties of genes predicted by your predictor that do not match an annotated gene? Write a paragraph for each question, supported by data and/or figures.

**Question 2. (20 points)**

A standard HMM is defined by state space *S,* an emission probability matrix *E*, and a transition probability matrix *T*, where $T(a,b) = \Pr[ P_{i+1} = b \mid P_i = a]$. HMMs operate under a first-order Markov model, i.e. $\Pr[ P_{i+1} = b \mid P_1 P_2 \ldots P_i ] = \Pr[ P_{i+1} = b \mid P_i ]$, where $P_i$ is the state at time step *i*.

In a second-order Markov model, $\Pr[ P_{i+1} = b \mid P_1 P_2 \ldots P_i ] = \Pr[ P_{i+1} = b \mid P_{i-1} P_i ]$. In other words, transition probabilities depend not only on the current state, but also on the previous state. In that case, one could define a transition probability table $T_2$, where
$T_2(a,b,c) = \Pr[ P_{i+1} = c \mid P_{i-1} = a, P_i = b ]$.
The initial state probability now needs to be defined for pairs of states:
$I_2(a,b) = \Pr[ P_1 = a, P_2 = b ]$.
Emission probabilities remain defined as previously: $E_2(a,y) = \Pr[ x_i = y \mid P_i = a ]$

a) (15 points) Consider a second order HMM $H_2$ with state space $S_2$, second-order transition table $T_2$, emission probability matrix $E_2$, and initial state probability distribution $I_2$. Describe a procedure to build a first-order HMM $H_1$ that is equivalent to $H_2$, i.e. such that for any sequence of observations *X* and path *P*, $\Pr_{H1}[ X, P ] = \Pr_{H2}[ X, P ]$.
b) (5 points) Demonstrate that $\Pr_{H1}[ X, P ] = \Pr_{H2}[ X, P ]$.

Good luck!